

The screenshot displays a Jupyter Notebook environment. On the left, a file explorer sidebar shows a directory structure with files: `..`, `.config`, `sample_data`, and `simpletestmatrix.txt`. The main area of the notebook contains a code cell with the following content:

Evaluate the fitness of a solution by using the Average Percentage of Faults Detected (APFD) metric

Your task is to complete this function

```
[8] # APFD Calculation
    # Takes a candidate solution and returns the fitness value based on APFD formula
    n = len(fault_matrix.index) # Number of test cases
    m = len(fault_matrix.columns) # Number of faults

    def fitness_function_APFD(candidate):
        # TODO: Complete the APFD fitness function
        faults_detected_set = set()
        total_tf_sum = 0

        for position, test_case_name in enumerate(candidate):
            test_row = fault_matrix.loc[test_case_name]
            detected_by_this_test = test_row[test_row == '1'].index
            for fault in detected_by_this_test:
                if fault not in faults_detected_set:
                    faults_detected_set.add(fault)
                    total_tf_sum += (position + 1)

        num_detected_faults = len(faults_detected_set)
        num_undetected_faults = m - num_detected_faults

        total_tf_sum += (num_undetected_faults * n)

    try:
        apfd = 1.0 - (total_tf_sum / (n * m)) + (1.0 / (2 * n))
    except ZeroDivisionError:
        return 0.0

    return apfd
```

[illegible]